

Priced In, Bit by Bit: Measuring Information Aggregation in Prediction Markets

Daniel Thomas*

Abstract. Prediction markets are typically evaluated on whether their prices are well calibrated forecasts of eventual outcomes. This paper asks a different question: how informative are prices? We use Shannon entropy and mutual information to quantify how much information observing prices tells us about the final outcome. Using a panel of roughly 50,000 closed binary markets on Polymarket, we find that observing prices on average resolves 53.65% of uncertainty surrounding outcomes, consistent across different market structures. Even over four months from resolution, prices reduce 36.36% of uncertainty, that figure jumping up to 90% one day from resolution. Finally, following Samuelson’s martingales, we show that while price changes are statistically predictable, any departure from a fair game is economically negligible, consistent with prices closely following an efficient information aggregation process. These findings are largely consistent with Hayek’s claim that prices aggregate dispersed information, with only small, though significant, departures from perfect efficiency.

Keywords: prediction markets, information aggregation, market efficiency, mutual information, entropy

JEL Classification: D83, G14, C13

1. Introduction

Despite their recent surge in popularity, prediction markets are not a novel idea. The earliest recorded example of a ‘prediction’ market goes all the way back to 1503, where people bet on the papal elections ([Rhode and Strumpf, 2013](#)). Five centuries later, traders on Polymarket wagered millions on who would ascend to the Papacy. Since ([Hayek, 1945](#)), economists have recognized that prices serve dual roles: they both allocate resources and aggregate information on the value of those resources. Prediction markets aim to exploit the information aggregating power of prices in an attempt to forecast future events, as argued by a wide range of leading economists ([Arrow et al., 2008](#)).

A price $p \in [0, 1]$ in a prediction market tells us how likely the market believes an outcome Y is. We’ll restrict ourselves to binary market outcomes only, so $Y \in \{0, 1\}$ (either the outcome doesn’t happen, or it does), with $Y = 1$ priced at p and therefore $Y = 0$ priced at $1 - p$. The equilibrium price p equals (or is very close to) the mean beliefs of the traders that the event will occur ([Wolfers and Zitzewitz, 2006](#)). The first modern academic prediction market was the University of Iowa’s Iowa Electronic Markets, introduced during the 1988 US presidential election. Although prediction markets contain many irrational or mistake-prone traders, empirical evidence from the Iowa Electronic Markets suggests that market prices remain highly informative because marginal traders responsible for price formation are substantially more rational than the average participant and arbitrage disciplines persistent pricing errors ([Oliven and Rietz, 2004](#)).

*St. Stephen’s College, University of Delhi. Email: danieljohannthomas@gmail.com. The code used in this paper is available on [GitHub](#).

Prediction markets have a distinctive property over other financial markets: they are one of the very few financial environments in which market efficiency can be evaluated directly because the hidden state of the world is eventually revealed (Thaler and Ziemba, 1988). Consider a stock X . Is the stock correctly priced today? There's no objective answer to that question. The stock's fundamental value is a theoretical object that remains unobservable forever. On the other hand, consider the event $Y = \text{"party } A \text{ wins the election."}$ Y starts out unknown, but eventually, nature will reveal $Y \in \{0, 1\}$ and thus, Y will be observed.

The fact that prediction market prices can be tested against an observed realization of outcomes fundamentally changes what can be tested empirically. In conventional financial markets, the Efficient Market Hypothesis (Fama, 1970) is typically evaluated only indirectly through implications such as return predictability, abnormal returns, or event studies, since the underlying fundamental value is never observed. Prediction markets instead allow direct evaluation of informational efficiency because realized outcomes become available after market resolution.

Existing evaluations of prediction markets have focused primarily on calibration, scoring rules, and forecast accuracy. While these establish whether prices are statistically well calibrated, they do not quantify how much uncertainty market prices eliminate about the eventual outcome. This paper adopts an information-theoretic perspective, measuring the reduction in uncertainty about the eventual outcome attributable to market prices through Shannon entropy and mutual information (Shannon, 1948). Hayek's central claim was that prices aggregate dispersed information. If prices aggregate information, then observing prices should progressively make us less uncertain about the future state of the world as we accumulate new information. Shannon defined information itself as a reduction in uncertainty produced by an observation, making Shannon entropy a natural metric to measure how much information prediction market prices contain.

This paper answers three questions regarding the informativeness of prices. Are they informative at all? If yes, how does the informativeness of prices change over time, and finally, how do prices become informative in the first place? First, we quantify how much uncertainty Polymarket prices resolve on average using mutual information. Second, we trace how this informativeness accumulates over time by measuring the mutual information across various time to resolution bins. Third, we use (Samuelson, 1965)'s martingales to test whether price changes can be predicted, which would negate efficient incorporation of information.

We find that prices resolve roughly 53.65% of outcome uncertainty on average, with this figure remaining consistent across Negative Risk markets too. Among markets that've been open for > 4 months, markets resolve 36.36% of outcome uncertainty even > 4 months from resolution, while this figure increases as time to resolution falls. We also see that while price changes can be predicted with statistical significance (as a result of the large sample size), any deviations from prices being a martingale are minuscule at best and cannot reliably be exploited for profit.

2. Methodology

2.1. Why information theory?

If we'd like to quantitatively study evidence for the Hayekian claim that prices aggregate dispersed information, a field named "Information Theory" is hardly a surprising choice. Hayek's

theory is fundamentally about communicating information. If prices can in fact communicate information, it should leave us better informed about the eventual realization of an event (which prediction markets have the unique feature of eventually observing). In simple terms, prices make us better informed, i.e. less uncertain about what will happen. Shannon's theory provides a natural framework because it gives us a way to quantitatively measure information as the *uncertainty reduced* regarding an outcome. This raises the empirical question of how we can quantify uncertainty and its reduction.

(Shannon, 1948) proved that if we have a set of possible events whose probabilities of occurrence are $p_1, \dots, p_n$, with the probabilities being the only thing we know about which event will occur, then the only measure of how much choice is involved in the selection of the event, i.e. how uncertain we are of the outcome, that satisfies a few mathematically desirable properties is of the form

$$H = -K \sum_{i=1}^n p_i \log p_i$$

with $K = 1$ being defined as the *entropy* of the set of probabilities $p_1, \dots, p_n$. Taking $\log_2$ gives entropy measure a unit he called "bits". Entropy is highest when both outcomes are equally likely, since the outcome is most difficult to predict, and falls as one outcome becomes increasingly certain. Thus, it provides an intuitive quantitative measure of uncertainty. Since Polymarket prices are just probabilities (in the binary case, just p_1 and p_2), we can define the entropy of an outcome Y on Polymarket to be $H(Y)$.

Economic logic tells us market prices provide additional information beyond the unconditional probability of an event. After observing the market price, uncertainty about the eventual outcome should decline if prices aggregate information successfully. We denote this as the *conditional entropy* of event Y given the market price using the notation $H(Y|P)$. Mathematically, $H(Y|P)$ is the average entropy of Y for each value of P , weighted by the probability of getting that particular value of P :

$$H(Y|P) = - \sum_p \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, P = p) \log_2 \mathbb{P}(Y = y|P = p)$$

Now that we have a measure for the unconditional uncertainty of Y and the conditional uncertainty upon observing the price, we can finally mathematically define information as uncertainty reduction. Shannon claimed "the rate of actual transmission [of information], R , would be obtained by subtracting from the rate of production (i.e., the entropy of the source) the average rate of conditional entropy." Since prices act as signals in prediction markets, the information transmitted by it is given by

$$I(Y; P) = H(Y) - H(Y|P)$$

Note that what Shannon defined as R , we'll stick to the modern convention and call this quantity *mutual information*, or just information for convenience. Mutual information will be our main object of interest in this paper. A large value tells us that observing prices dropped uncertainty by a large amount relative to the uncertainty of the unconditional distribution of outcomes, thus, larger values of mutual information are more desirable.

2.2. Information vs calibration

There have been numerous studies that have concluded that besides a few behavioral biases (like the favorite longshot bias), prediction markets are generally accurate and well calibrated, often outperforming professional forecasters and polls in a variety of statistical tests. Studies that show this have been summarized in (Snowberg et al., 2012). Our objective, however, is different. Rather than asking whether prices are *accurate* forecasts, we ask how *informative* they are. Consumers of economic forecasts care not just about the accuracy of the mean prediction, but the uncertainty attached to it as well. The standard proxy for uncertainty is disagreement among forecasters. As (Snowberg et al., 2012) show, that is a poor proxy for actual uncertainty. Using information theory, we provide an alternative way to measure uncertainty in markets.

The difference between information and calibration is one that has real implications; it isn't just a semantic difference. Consider a market for an election between two equally likely candidates A and B that prices candidate A at 0.5. Observing the price hasn't given us any new information, as the unconditional distribution of outcomes is roughly 50% anyways. Suppose a major news story breaks out and pushes candidate A 's price to 0.95. This gives us a lot of new information, as we now know that the market thinks A is almost certain to win. Notice how the informativeness of a market can be established before any calibration metric can be analyzed. Calibration evaluates the quality of a prediction *after* the uncertainty has been resolved. Information measures how much of that uncertainty was eliminated *while* the outcome was still unknown. These are complementary, but fundamentally different, dimensions of market quality.

3. Data

3.1. Source and Construction

Our dataset combines information from Polymarket's Gamma API, which gives us cross-sectional market data, and the Polymarket CLOB (Central Limit Order Book), which gives us hourly snapshots of the price history across our markets.

To calculate information-theoretic quantities, we need a well defined random variable whose realizations are eventually observed. This restricts our sample to closed markets only. We further restrict our sample to closed binary markets. This keeps the interpretation simple as for an outcome Y , $p = P(Y = 1)$ and $1 - p = P(Y = 0)$. Interpretation gets more complex as we move on to multi-outcome markets and keeps our analysis a lot cleaner. As these markets are already resolved, there is no trading activity on them and hence the Gamma API reports a liquidity of zero on almost all these markets. Hence, we do not use liquidity in our analysis.

The initial Gamma dataset contained 603,409 binary resolved markets, with data ranging from the start of Polymarket's Gamma data availability till July 18th, 2026. We then sort the data by volume and only select the top 50,000 markets by volume (roughly 71.5th percentile) to be part of our sample. This ensures that our analysis is not skewed by uninformative low volume markets. Although we started out with 50,000 observations, we had to drop 14 observations with no volume, 3 observations without a specified market close time and a further 12 observations that did not have a price of 0 or 1, i.e. did not cleanly resolve, leaving us with 49,971 markets in our sample. The market IDs of the sample are then matched to the price history data from CLOB to get a panel with historical price data for each market in our sample. This process leaves

us with a panel that is indexed by market ID and timestamp (hourly) and has 43.4 million rows observations corresponding to 49,971 markets.

For each observation, the remaining time until market resolution is computed as the difference between the observation timestamp and the market closing time. This variable serves as the primary temporal dimension of the analysis. To facilitate comparisons across markets with widely differing lifetimes, time to resolution is sorted into equal length bins. Prices are also binned into equally spaced 5 percentage point intervals to estimate how information measures vary across different predicted outcome probabilities.

3.2. Summary Statistics

Table 1: Market Summary Statistics

Variable	Mean	Median	SD	Min	Max
Trading volume (\$ million)	1.126	0.160	10.984	0.052	1531.479
Market duration (days)	38.727	13.385	64.309	0.088	939.485
Resolved Yes (=1)	0.286	N/A	N/A	N/A	N/A
Negative Risk (=1)	0.624	N/A	N/A	N/A	N/A

Table 2: Panel Summary Statistics

Variable	Mean	Median	SD	Min	Max
Price	0.203	0.070	0.270	0	1
Time to resolution (days)	18.77	6.522	31.034	0.007	294.957

4. Question 1: Are prices informative?

The methodology section has given us a quick primer on a quantitative definition of information. Now it is time to apply that and answer the central empirical question of this paper: how informative are these prediction markets?

To answer this, we must first calculate the baseline entropy of outcomes, i.e. how unpredictable are outcomes in binary markets on Polymarket on average? Theoretically, a binary outcome has entropy $H(Y) = -(p \log_2(p) + (1 - p) \log_2(1 - p))$. This expression is maximized at $p = \frac{1}{2}$, and thus we find that the maximum possible entropy a binary outcome can have is

$$H(Y) = 1 \text{ bit}$$

Running the calculations on our dataset shows us that binary events on Polymarket have a baseline entropy of 0.8635 bits. This tells us that Polymarket binary events have roughly 86% of the maximum possible uncertainty for a binary event. Simply put, without looking at prices, market outcomes are highly unpredictable.

Now we introduce prices into the picture. To calculate $H(Y|P)$ precisely, we'd need to estimate $H(Y)$ for every possible price value, which is not practical since price is a continuous variable here. Hence, we've discretized price into 20 equally sized bins, each bin being of length 0.05. This lets us calculate $H(Y)$ within each price bin and then take the average of $H(Y)$, weighted by the relative frequency of each price bin to get $H(Y|P)$. Since we've used discrete bins, this is only an approximation of $H(Y|P)$, but a generally accurate approximation nonetheless. However, we must note that our values of $H(Y|P)$ are thus sensitive to changes in the price bins. With this setup, we obtain $H(Y|P) = 0.4002$ bits. While the eventual outcome obviously remains uncertain, much of the original uncertainty has already been resolved before the market even closes.

The difference between these quantities gives us $I(Y; P) = 0.4633$ bits. Expressed as a percentage of the original uncertainty, we see that prices alone have reduced the uncertainty regarding outcomes by 53.65%. This figure tells us observing market prices resolves more than half of the uncertainty surrounding future outcomes before those outcomes are even realized. This provides empirical evidence that prediction markets contain substantial information about the events they are designed to forecast.

4.1. Negative Risk Markets

Negative risk is a market structure where a combination of positions can reduce risk to zero and potentially create a guaranteed profit by exploiting mispricings. Polymarket has many such markets, usually when dealing with a set of mutually exclusive outcomes. For example, we could have three separate markets for the same award: movie $A/B/C$ wins the Oscar for Best Picture, with each of the three markets being a binary yes/no market for each movie. Assuming the Academy only awards the title to one movie, which it always has done, then one and only one of these 3 markets will resolve to a yes. If you buy a yes contract on all 3 markets, you are guaranteed a payoff of a dollar from the market that resolves yes, while gaining nothing from those that resolved no.

It is important for Polymarket to keep track of markets tied by negative risk due to the presence of arbitrage. If the cost of a yes contract on all 3 markets is less than a dollar, you've got a free profit by buying all three yes contracts. Polymarket has a mechanism to convert no shares into yes shares across such markets, allowing arbitrage to push yes prices towards summing up to 1. This presents a problem for our information aggregation story as now prices can change due to arbitrage, even without the introduction of new information into the market.

Our main dataset has 31,190 negative risk markets and 18,234 markets without negative risk. To run this robustness check, we first filtered our sample down to only closed binary markets as before. Then, we created two separate datasets that comprise entirely of negative risk and no negative risk markets. Finally, we picked the top 50,000 markets by volume for each dataset, giving us 3 separate datasets in total. From this, we get the results:

Table 3: Entropy results

Market type	Entropy	Conditional entropy	Information	% uncertainty reduced
Combined	0.86347	0.40019	0.46328	53.654%
Negative Risk	0.78834	0.37290	0.41544	52.698%
No Negative Risk	0.94988	0.45245	0.49743	52.367%

Immediately, the differences in $H(Y)$ across the three groups stand out. Markets that don't have negative risk have an unconditional entropy that is 95% of the maximum possible entropy, making them highly uncertain. Part of the reason why is likely because negative risk markets generally have a few clear frontrunners, while other options are extremely unlikely. Back to our Oscars example, there are typically very few movies that are considered genuine contenders for Best Picture. If we look at the markets for longshots, they're probably priced very low (< 0.05) and will probably resolve to no. Indeed, the data shows that markets with negative risk resolve no much more often. This is one possible reason that contributes to lower baseline entropy, although there may be other reasons as well.

Table 4: Percent of markets that resolve 'Yes'

Market type	Percent
Combined (NegRisk)	23.567
Negative Risk	23.027
Combined (no NegRisk)	36.895
No Negative Risk	39.012

What's remarkable is that although the three different datasets are composed of different markets with different negative risk structures and with different rates of resolving yes (different rates of $P(Y = 1)$, which is directly used in entropy calculations), market prices eliminate the same percentage of unconditional entropy in all 3 markets. The presence of negative risk arbitrage affects the amount of uncertainty present in a market, but it does not appear to affect the proportion of that uncertainty incorporated into prices. This points to the underlying information aggregation process being quite stable across various market designs.

5. Question 2: The dynamics of informativeness

The previous section established that market prices eliminate over half of the uncertainty surrounding future outcomes on average. However, an average alone tells us little about when this information enters prices. It could come in months in advance, or when an event happens that makes the outcome obvious. Prediction markets telling us that an event will happen after the general consensus has ruled it as extremely likely is not very informative. An informative market must certainly have the ability to be informative long before the event is considered "settled".

Before we plot entropy or information against time, we must be mindful a potential selection bias: markets have different lifespans. All closed markets must, by definition of being closed, have existed an hour before resolution, while not all markets need to be 4+ months old. Therefore,

directly plotting would give us a different sample size in each time to resolution bin. To control for this, we create ‘cohorts’ of markets that last roughly the same amount of time. Now, plotting entropy or information against time for a given cohort would give us the same sample size across every bin, dealing with selection bias issue. We plot the > 4 months to resolution cohort here ($n = 4618$), as it gives us the longest timespan to analyze informativeness over time. The results for the other cohorts are largely similar. The plots for other cohorts can be viewed on the [GitHub](#).

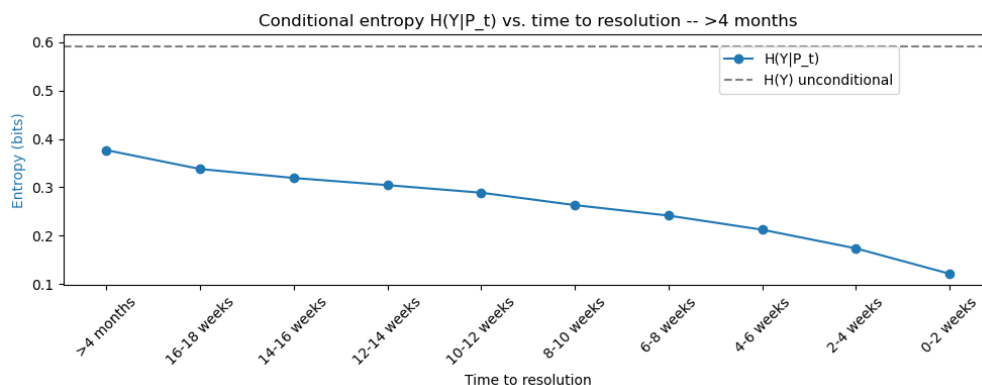


Figure 1: Conditional entropy over time.

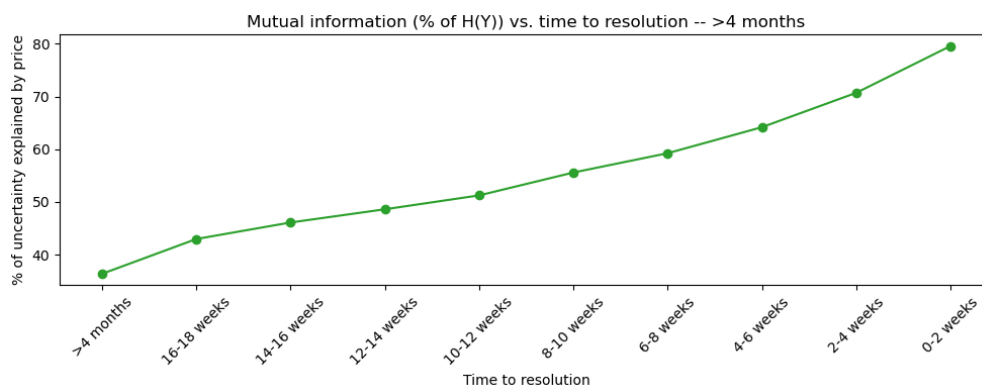


Figure 2: Mutual information as a percentage of unconditional entropy.

Note that the two figures show the same phenomenon from complementary perspectives. For each cohort, we first compute the unconditional entropy of market outcomes, $H(Y)$, which is a constant across all time to resolution bins since the distribution of Y is fixed for a given cohort. This is the baseline from which we subtract $H(Y|P)$ for each time to resolution bin to get the informativeness of prices over time on Polymarket. Notably, even markets that are over four months from resolution have already resolved 36.36% of the uncertainty of outcomes. Information aggregation is clearly a continuous process, one that starts long before outcomes become obvious. With 0-2 weeks to resolution, the market prices resolve an incredible 79.59% of uncertainty. If we refine the bins to use 0-1 days as the last time to resolution bin, that figure jumps up to 90.01%.

5.1. Rate of information aggregation

From the above figure 2, we see that the informativeness of the market measured in mutual information terms is monotonically increasing in our sample. While all the bins are evenly spaced, the > 4 months bin is open ended. This was unavoidable due to the presence of some very long lived markets. Besides that bin, the rest looks to be increasing at roughly an increasing rate, with the increase in informativeness being greatest closest to resolution. Plotting the first difference of mutual information as a percent of unconditional entropy, we get:

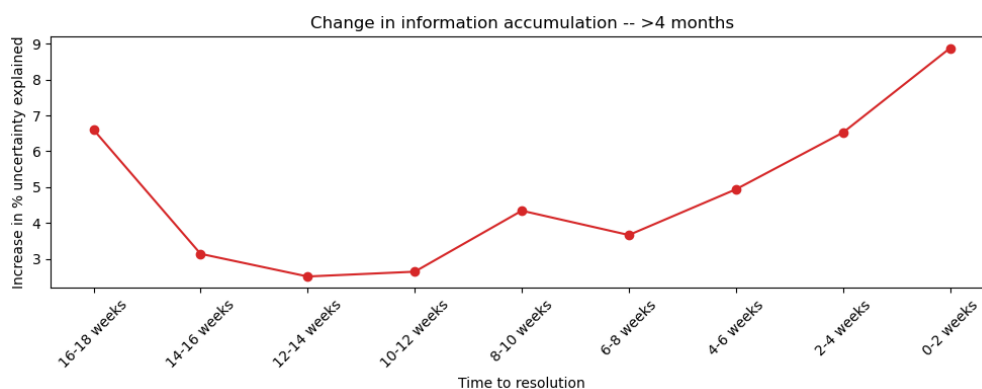


Figure 3: Change in % uncertainty reduced over time

The 16-18 week bin is skewed since its change relies on the open ended > 4 month bin, the only one that isn't evenly spaced. The rest presents a roughly increasing rate of information aggregation, with a slight dip 6-8 weeks before resolution, before continuing to increase. The (mostly) increasing first discrete derivative suggests that the information accumulation over time on Polymarket is a convex function, which would make sense given that there's simply more information out there for the market to absorb as we draw closer to resolution.

These results show that information aggregation in prediction markets is on average a gradual process, not confined to the moments right before resolution. Markets do contain substantial amounts of information many months before resolution. The average informativeness reported in the previous section is therefore not driven solely by prices immediately before market resolution but reflects information that has already accumulated throughout the market's lifetime. The results also suggest that the price's stock of information increasing at an increasing rate, although this is hard to definitively establish using just first differences of a discrete approximation.

6. Question 3: How do prices incorporate information?

So far, we've used mutual information to show that prices contain substantive information and how that information content increases over time. However, a natural question remains: how does information enter these prices? There are numerous plausible mechanisms. Perhaps news is immediately efficiently incorporated into prices by rational traders. Maybe information is gradually diffused into the market over time. It is also possible that traders may initially overreact to new information, leading to the price overshooting and then returning to an equilibrium. Across all three mechanisms, we could end up with very different price dynamics but similar information trajectories.

One way we can distinguish between these cases is by looking at serial correlation in price changes. Following (Fama, 1970), if prices immediately react to and incorporate new information, then you wouldn't expect any persistent pattern in price changes. Changes in price would show zero serial covariance (and thus no autocorrelation). On the other hand, gradual diffusion of positive news would lead to a persistent upward pattern in price changes, and overreaction would lead to a correction that shows persistent downward price changes, and vice versa for negative news.

For lags $k \in \{1, \dots, 24\}$, we compute the correlation between the current price change Δp_t and the k -hour lagged price change Δp_{t-k} per market. This gives us the following plot for autocorrelation:

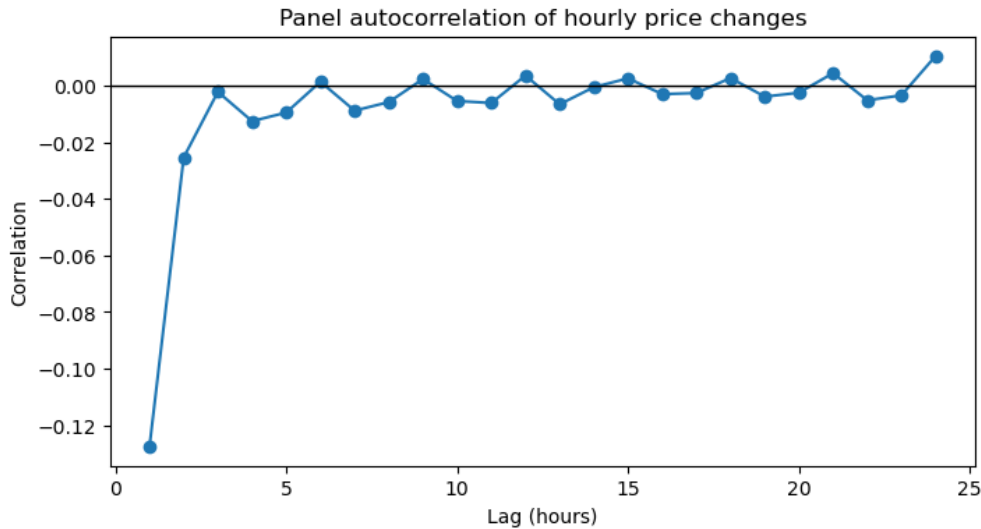


Figure 4: Panel autocorrelation of hourly price changes.

We observe slight negative autocorrelation at the one hour gap, after which any autocorrelation, although statistically significant, is economically negligible and almost certainly not exploitable for profit. The initial negative autocorrelation could be due to market microstructure effects (like bid-ask bounce, although Polymarket's documentation doesn't make it clear whether prices are last trade price or the bid-ask midpoint). Inefficiencies as the cause cannot be entirely ruled out, however, they are corrected in about an hour, so any possible inefficiency fails to persist.

It could be argued that our autocorrelation plot is an artifact of aggregation, where we genuinely have different types of markets that incorporate information differently, yet aggregating the results smoothens differences and presents it as looking like an informationally efficient market. To test this, we plot the coefficients β_1, β_2 and β_3 obtained from regressing Δp_t on $\Delta p_{t-1}, \Delta p_{t-2}$ and Δp_{t-3} for each market and plot the distribution of β s obtained. We see no signs of multimodal distributions here. This tells us that autocorrelation patterns observed are pervasive and consistent across markets, rather than an artifact of pooling heterogeneous markets. Combined with the very small autocorrelation observed beyond one hour, this points towards markets efficiently incorporating information.

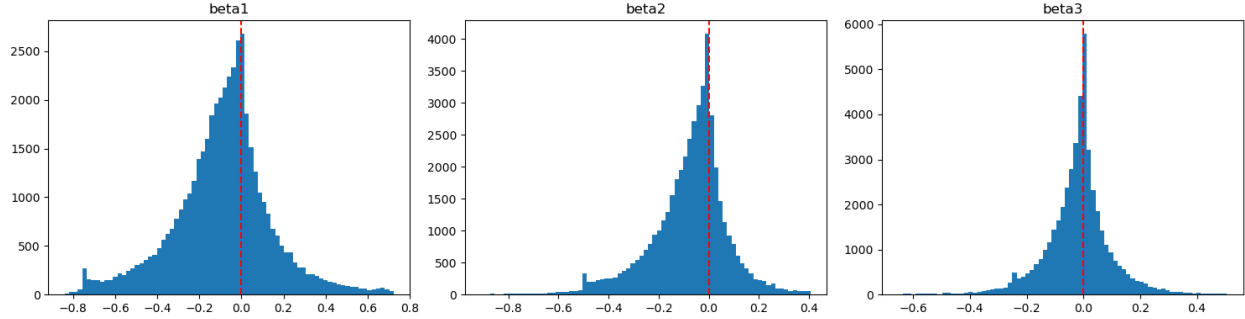


Figure 5: Distribution of autocorrelation coefficients.

6.1. Polymarket prices as a fair game

So far, we've tested whether a market's own price changes can predict future price changes, i.e. Fama's weak form efficiency. A broader result is (Samuelson, 1965), who showed that if prices are formed as the conditional expectation of an outcome given *all available information*, $p_t = \mathbb{E}[Y|I_t]$, then price changes behave like a fair game wrt the information set, i.e. $\mathbb{E}[\Delta p_t | I_t] = 0$: given everything we know, the expected future price equals today's price. Prices can only change if new information arrives, and by definition that cannot be predicted. This asserts that no piece of information can predict future price changes.

Unfortunately, we do not have access to every single possible datapoint players in the market may use to make their decisions. However, we can expand our previous tests to use a wider information set, one that includes the current price, time to resolution, and volatility. Regressing Δp_t on our available information, we get the OLS result:

Table 5: Predictability of Hourly Price Changes

Variable	Coefficient	Std. Error
Intercept	-0.0009***	0.00001
Price (p_{t-1})	0.0043***	0.00007
Time to Resolution (std.)	0.00002***	0.00001
Δp_{t-1} (std.)	-0.0027***	0.00004
Δp_{t-2} (std.)	-0.0010***	0.00002
Δp_{t-3} (std.)	-0.0003***	0.00002
Volatility (std.)	-0.0005***	0.00001
Observations	43,001,560	
R^2	0.025	
F-statistic	1387	
Clustered SEs	Market	

Notes: Dependent variable is the hourly price change, Δp_t . Variables marked (std.) are standardized to have mean zero and unit variance. Volatility calculated over a 24 hour rolling window. *** $p < 0.01$.

43 million observations effectively makes any result statistically significant. What matters here is economic significance. Notice that even when conditioning on a much wider information set, our model can only explain 2.5% of variation in hourly prices ($R^2 = 0.025$), with every

variable having very small coefficients. This tells us that while we can predict price changes with statistical significance, in practice, the magnitude of that predictability is negligible at best. The estimated effects are small enough that they're unlikely to support profitable strategies once transaction costs and other frictions are taken into account. Consistent with Samuelson's fair game hypothesis, publicly available past information provides little practical ability to forecast subsequent price movements on Polymarket.

7. Conclusion

This paper set out to answer three questions: whether prediction market prices are informative, how that informativeness evolves over time, and whether the process by which it accumulates looks efficient. The answers to these three questions build up on each other and form a clear narrative. Prices resolve roughly half of outcome uncertainty on average, with informativeness climbing from resolving around a third of the total uncertainty at long horizons to nearly all of it in the final day before resolution. The martingale results speak to what that remaining, unresolved uncertainty is likely *not*: our tests suggest that whatever information is already reflected in recent prices, volatility, and time to resolution is being incorporated efficiently, since none of it meaningfully predicts future price movements. What we cannot say is whether the residual uncertainty at any given point is genuinely undetermined, or whether it reflects dispersed information that simply hasn't reached the market yet, as price unpredictability is consistent with both. We also have not yet formally tested how sensitive our mutual information estimates are to the choice of price bin width; both are natural next steps.

In general, this paper's contribution is less about Polymarket specifically and more about a method. In markets where a latent variable that drives pricing is eventually observable (the outcome happening or not here), entropy and mutual information offer a way to ask how much a price tells you, independent of whether that price turns out to be 'right'. As prediction markets are increasingly proposed as forecasting tools for policy and institutional decision making, we believe the distinction between a market being accurate and a market being informative is one worth taking seriously. For policymakers and other users of forecasts, this distinction matters because a market can provide useful information long before its forecast can be judged correct. Decisions have to be made before outcomes are known, so what a market tells us today can matter even when its final forecast cannot yet be evaluated.

References

- Kenneth J. Arrow, Robert Forsythe, Michael Gorham, Robert Hahn, Robin Hanson, John O. Ledyard, Saul Levmore, Robert Litan, Paul Milgrom, Forrest D. Nelson, George R. Neumann, Marco Ottaviani, Thomas C. Schelling, Robert J. Shiller, Vernon L. Smith, Erik Snowberg, Cass R. Sunstein, Paul C. Tetlock, Philip E. Tetlock, Hal R. Varian, Justin Wolfers, and Eric Zitzewitz. [The Promise of Prediction Markets](#). *Science*, 320(5878):877–878, 2008.
- Eugene F. Fama. [Efficient capital markets: A review of theory and empirical work](#). *The Journal of Finance*, 25(2):383–417, 1970.
- Friedrich A. Hayek. [The Use of Knowledge in Society](#). *The American Economic Review*, 35(4): 519–530, 1945.

- Kenneth Oliven and Thomas A. Rietz. [Suckers Are Born but Markets Are Made: Individual Rationality, Arbitrage, and Market Efficiency on an Electronic Futures Market](#). *Management Science*, 50(3):336–351, 2004.
- Paul W. Rhode and Koleman Strumpf. [The long history of political betting markets: An international perspective](#). In Leighton Vaughan Williams and Donald S. Siegel, editors, *The Oxford Handbook of the Economics of Gambling*, pages 560–583. Oxford University Press, 2013.
- Paul A. Samuelson. [Proof That Properly Anticipated Prices Fluctuate Randomly](#). *Industrial Management Review*, 6(2):41–49, 1965.
- Claude E. Shannon. [A mathematical theory of communication](#). *The Bell System Technical Journal*, 27(3):379–423, 1948.
- Erik Snowberg, Justin Wolfers, and Eric Zitzewitz. [Prediction Markets for Economic Forecasting](#). Discussion Paper 6720, Institute for the Study of Labor (IZA), 2012.
- Richard H. Thaler and William T. Ziemba. [Anomalies: Parimutuel betting markets: Racetracks and lotteries](#). *Journal of Economic Perspectives*, 2(2):161–174, 1988.
- Justin Wolfers and Eric Zitzewitz. [Interpreting prediction market prices as probabilities](#). Working Paper 06-11, Rodney L. White Center for Financial Research, The Wharton School, University of Pennsylvania, 2006.